

Introduction to Econometrics Final Project

Dr. Soliman
Clemson University
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Guidelines and Advice

Purpose. The final project evaluates whether you can ask a clear empirical question, find appropriate data, apply econometric tools from class correctly, interpret your results, and explain the limitations of your analysis. Original topics are welcome, but a successful project does not require a novel research design. It requires a well-defined question, suitable data, transparent methods, and careful interpretation.

1. Start Here: Choosing a Project Path

You may choose one of two paths. Both can earn full credit. The key is to be honest about what your design can and cannot show.

Path	Best suited for	What your paper must do
Path A: Correlational project	Questions about relationships, prediction, or descriptive patterns. This path is usually more feasible when treatment timing or a credible policy shock is unavailable.	Estimate and interpret associations carefully. Do not claim causality. In the limitations section, explain what would be needed to estimate a causal effect.
Path B: Causal project	Questions about the effect of a policy, program, shock, or intervention. This path is harder because it requires a credible comparison group and a plausible identification strategy.	State the causal parameter, motivate the identification strategy, discuss the identifying assumptions, and explain why the comparison group is informative.

Related examples. The same topic can often be framed as either correlational or causal:

Topic	Correlational question	Causal question
Marijuana and crime	What is the relationship between marijuana use and violent crime rates across states?	What is the effect of legalizing recreational marijuana on violent crime?
Minimum wage and employment	Are states with higher minimum wages associated with lower teenage employment?	What is the effect of a minimum wage increase on teenage employment?
Police staffing and crime	Do cities with more police officers per capita have lower crime rates?	What is the effect of a police staffing increase on crime?
Unemployment and opioid mortality	Are counties with higher unemployment rates more likely to have higher opioid mortality?	What is the effect of a plant closure or local labor demand shock on opioid mortality?

2. Practical Advice

Choose the data before committing to the question

- Your primary constraint is data availability. A good question that cannot be answered with available data is not a good final project topic for this course.
- Before committing, verify that you can download the relevant data in a usable format, such as CSV, Excel, or a clean fixed-width file. Do not build the project around data that only appear in PDFs.
- You should aim to find a dataset with at least 1000 observations. Furthermore, make sure the data contain enough variation to answer your question. For a causal project, you usually need variation across units and over time, such as states adopting a policy in different years. Projects using substantially smaller datasets require instructor approval.

Match the method to the question

- For a correlational project, OLS, multiple regression, fixed effects, and sensitivity checks may be sufficient if interpreted correctly.
- For a causal project, you need a credible source of identifying variation, such as a policy change, threshold, shock, or staggered adoption across places.
- Using advanced methods is not necessary. Correctly applying and explaining methods from class is sufficient for a strong paper.

Write narrowly

- Do not report every regression you estimated. Report the specifications that answer your question and help the reader understand the evidence.
- Use clear variable names, readable tables, labeled figures, and captions that can be understood without reading the full text first.

3. Format and Submission Requirements

- Write the paper in R Markdown and submit the PDF generated from your Markdown file.
- Do not show R code in the PDF. The final paper should read as a crisp, standalone document.
- The expected length is about 5 pages of text, not counting tables, figures, references, or appendices. Papers below 3 pages of text are usually too thin; papers above 8 pages of text are usually too long.
- The project is worth 250 points. The required sections and point values are listed below.

4. Required Paper Sections

Use the section headings below in your final paper. You may add subsections as needed.

1. Literature Review (30 points)

Use prior work by economists or other social scientists to motivate your question, guide your variable choices, and clarify what mechanisms or confounders may matter.

- Requirement: cite at least 3 academic papers. A comprehensive literature review is not expected.
- A good literature review briefly explains what the prior work teaches you and how it informs your empirical choices.

2. Data and Summary Statistics (60 points)

Describe the data, where you obtained it, the unit of observation, the sample period, the key variables, and the main strengths and limitations. Demonstrate that you understand what the data can and cannot measure.

- Requirement: include at least 5 clearly labeled figures.
- Requirement: use at least 3 different types of visualizations, such as a line chart, bar graph, histogram, scatter plot, or map.
- All figures must have readable axis labels and captions. Do not use raw variable names such as `bthyr` when the axis should say Birth Year.

3. Methodology (30 points)

Describe and motivate the econometric techniques used to answer your question. The goal is to make the main analysis replicable without listing every coding decision.

- Requirement: include at least 1 and no more than 3 estimating equations.
- Fully define the outcome, key explanatory variable, controls, fixed effects (if used), error term, and the parameter of interest.
- For a causal project, explicitly state the identifying assumption. For a correlational project, explicitly state that the coefficient should not be interpreted causally.

4. Results and Analysis (60 points)

Present and interpret your main empirical results. Do not simply report coefficients. Explain their magnitude, sign, statistical significance, and economic or substantive meaning.

- Tie the results back to the research question.
- Discuss whether the estimates are large or small in context.
- Explain how the results change, or do not change, across your main specifications.

5. Discussion and Limitations (70 points)

Discuss the main limitations of your data, empirical strategy, and interpretation. Every paper has limitations, including omitted variable bias, selection, reverse causality, measurement error, external validity, or identifying assumptions.

- Everyone: include at least 2-3 paragraphs discussing the limitations of your current study.
- If you choose Path A, the correlational path: include at least 2-3 additional paragraphs explaining how one could estimate a causal effect. This can be a hypothetical experiment or a real-world policy design. Be specific about the treatment, comparison group, outcome, and identifying variation.
- If you choose Path B, the causal path: explain the identifying assumptions and discuss where they may fail in your setting.

5. Final Checklist

- The research question is clear and appears early in the paper.
- The paper makes clear whether it is taking Path A or Path B.
- The data are downloadable and appropriate for the question.
- The paper uses methods from class correctly.
- The tables and figures are readable without relying on raw variable names.
- The interpretation distinguishes correlation from causation.
- The limitations section is specific rather than generic.
- The PDF does not display R code.

Two Notes: 1) It is possible to write an excellent paper without following every piece of advice above, but the required sections and submission format still apply. 2) The presentation is worth almost as much as the paper, and happens before submission of the final project so plan your time accordingly.